

Hypothetically, can we use Ai to generate energy?

Posted by u/Linorelai • 0 points • 24 comments

The servers generate a concerning amount of heat, as far as know. Can we use it to boil water to generate electricity to partly replace oil and thus make the environmental impact even?

Comments

u/No-Flatworm-9993 • 1 points • 2025-12-13 22:08 UTC

The cooling systems do in fact boil water. But if you tried to used that energy to do work, it would be a slower, worse cooling system.

u/rubysundance • 1 points • 2025-12-13 19:27 UTC

I'm hoping when AI becomes fully functional it's used for developing a new type of energy that our human minds couldn't figure out.

u/truncated_buttfu • 2 points • 2025-12-13 19:15 UTC

We are doing exactly that in some places. I know a super computer center that uses its excess heat to heat up warm water for the district heating system. This is a very nice scheme since the heat doesn't necessarily need to *boil* the water, as long as it makes it warmer it's a win. And they put a hot water tank very close to the server hall to minimize heat loss in the pipes between.

However, this doesn't make it "even", just a bit less wasteful. It's hard to capture heat from the air and transfer it in an efficient way, a lot of the heat will inevitably get lost in the system.

u/green_meklar • 2 points • 2025-12-13 19:09 UTC

The heat doesn't appear by magic, it comes from electricity. If not used for servers, that electricity would be available for other things.

Can we reuse the heat? Yes, and we sometimes do. I remember that in Europe there was (and perhaps still is) an effort to put server racks in people's homes where the heat coming from them would offset the home heat bill.

Can we use the heat to boil water? Sort of, but it's harder than it sounds because the server electronics don't work fast or reliably at temperatures as high as 100C. They're typically designed for ambient temperatures of something like 30C or lower. Yes, it's possible to *concentrate* heat using air-conditioning techniques, but it requires a lot of extra expense and infrastructure and often isn't worth it just to recover something as easily acquired as heat for boiling water.

Can we recover the heat for power even at temperatures below 100C? Yes, we could use it to expand air and drive an air turbine. However, putting a turbine in the way of the escaping hot air makes it that much harder to push the air out. Normally, we're already doing extra work to push the hot air away from the servers and pull new cold air in, so putting an air turbine in the way would probably be a net power loss.

u/generic_redditor_71 • 4 points • 2025-12-13 19:07 UTC

Electrical generation from heat gets more efficient the higher the temperature of the heat source. Steam turbines at power plants run at hundreds of degrees Celsius. Computers need to stay well below 100 C. That's not hot enough to generate a useful amount of electricity. The best you could do is use the heat for heating rooms and providing hot water to nearby buildings, and some such systems exist, but most of the time it's not worth the cost and complexity.

u/JustSomeGuy_56 • 1 points • 2025-12-13 19:06 UTC

Circa 1980 I worked for an insurance company. When we built a new building we used the cooling water from our big IBM mainframe computers to heat it. We got some award from the Dept of Energy but it wasn't all that effective and the added expense more than offset the savings

u/SpaceCancer0 • 1 points • 2025-12-13 19:03 UTC

Yes but no. The gains would be so little that you're better off optimizing your power plants

u/katarh • 3 points • 2025-12-13 19:03 UTC

No. The temperature water boils is 100C. That temperature is enough to outright cook electronic components, whose max safe operating temperature is usually between 60-90C.

This is why water is used to *cool* thermal components in electronics, because it is a very efficient carrier of heat and will keep the electronics below the dangerous temperatures without itself starting to expand into a gas.

u/DavidRellim • 12 points • 2025-12-13 18:59 UTC

...a staggering lack of respect for the laws of entropy here

u/MikeKrombopulos • 2 points • 2025-12-13 19:04 UTC

They're talking about reclaiming some energy from the waste heat, not generating a net positive amount of energy.

u/DavidRellim • 1 points • 2025-12-13 19:10 UTC

thorreally?facememe

u/MightyMeepleMaster • 5 points • 2025-12-13 19:06 UTC

Sorry mate, but OP wrote "generate energy".

Case closed.

u/laz1b01 • 2 points • 2025-12-13 22:12 UTC

Wouldn't the word "generate energy" be appropriate for a person that doesn't know much about physics? Technically, you are generating energy - it just so happens the energy consumed by AI is much greater than the energy you're able to generate from the heat of the computing hardware.

u/drowning35789 • 1 points • 2025-12-13 18:55 UTC

It would probably take more energy to run the ai

u/ExtraDependent883 • -3 points • 2025-12-13 18:54 UTC

I use the heat from my soiled DEPENDS®™ For men™ crotch guard stay dry pads to warm the other half of my ham and cheese when I'm at a remote and chilly job site...very efficient highly sustainable ebergbybmanagment

u/BackFromMyBan2 • 2 points • 2025-12-13 19:01 UTC

Thanks, I will consider this product the next time I shop at my local retail store!

u/ExtraDependent883 • 1 points • 2025-12-14 05:57 UTC

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u/ExtraDependent883 • 1 points • 2025-12-14 01:47 UTC

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u/Childoftheway • 5 points • 2025-12-13 18:48 UTC

That would be cool if AI was powered by humans on treadmills. You'd have to work for your answers to your own and society's benefit.

u/No-Flatworm-9993 • 1 points • 2025-12-13 22:09 UTC

It would be easier to pedal a bicycle to hit me in the head with a hammer, if the goal is being dumber

u/AgentElman • -16 points • 2025-12-13 18:45 UTC

We can use the heat from servers to heat things. Some places are starting to do that.

But AI uses very little energy compared to everything else. The complaints about AI energy are anti-AI hysteria, not a practical issue.

AI uses less energy than the rest of the internet such as streaming services. People are only complaining about it because it is AI.

u/That_1-Guy_- • 10 points • 2025-12-13 18:52 UTC

I mean we can literally see Electricity prices rise as data centers are being built, it's not just random rhetoric it's observable

u/GreatNameLOL69 • -6 points • 2025-12-13 18:48 UTC

Yeah, I know some people who literally go from "this painting is peak, who made it?" to "this is utter trash" in a matter of seconds.

u/mugenhunt • 15 points • 2025-12-13 18:44 UTC

While you could recover some of the energy wasted on AI that way, it wouldn't be very efficient.